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SUPERSEDING
MIL-J-26014A (USAF)
3 June 1957

MILITARY SPECIFICATION

JACK, HAND, HYDRAULIC, AXLE, 20 TON CAPACITY, TYPE F-1

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of dolly-mounted, hydraulic, 20-ton-capacity axle jack, designated Type F-1.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-601	Boxes, Wood, Cleated Plywood

MILITARY

MIL-P-116	Preservation, Methods of
MIL-H-5606	Hydraulic Fluid, Petroleum Base, Aircraft and Ordnance
MIL-H-81019	Hydraulic Fluid, Petroleum Base, Ultra-Low Temperature

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: San Antonio ALC/MMEDO, Kelly AFB, Texas 78241 by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1730

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	MIL-STD-130	Identification Marking of U. S. Military Property
*	MIL-STD-831	Test Reports, Preparation of
*	MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking and Waterproofing; with Appropriate Test Methods

DRAWING

Air Force

53D22034	Jack - Hand, Hydraulic Axle, 20 Ton Capacity, Type F-1, Assembly of

PUBLICATION

Air Force-Navy Aeronautical Bulletin

405	Storage Life - Aeronautical Articles

* (Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.1.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

* 3.1 First Article. This specification makes provisions for first article testing.

3.2 General. The requirements specified on Drawing 53D22034 are applicable as requirements of this specification. When the requirements of the drawing and this specification conflict, the requirements of this specification shall govern.

3.3 PERFORMANCE

3.3.1 Static proof. The jack shall be capable of supporting and raising a 30-ton load vertically. The jack shall be capable of withstanding a horizontal load of 3 tons while supporting a 20-ton vertical load in the fully extended position.

3.3.2 Low temperature. The jack shall be capable of satisfactory operation at -54° C.

3.3.3 Life. The jack shall be capable of satisfactory operation after 200 cycles of raising a 20-ton load vertically, and 50 cycles of raising a 20-ton load vertically while being subjected to a horizontal load of 3 tons.

3.3.4 Leakage. The cylinder and ram assembly shall not leak when subjected to a hydraulic pressure of 7,500 psi.

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3.3.5 By-pass valve. The by-pass valve shall be adjusted to the valve specified on the drawing.

3.4 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

* 3.5 Hydraulic fluid. Hydraulic fluid conforming to MIL-H-5606 shall be utilized.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of tests. The inspection and testing of the jack shall be classified as follows:

a. Acceptance test . . . . . See 4.2

* b. First Article testing . . . . . See 4.4

* 4.1.2 First Article testing is required to determine comformance of the jack to the drawing with respect to materials, workmanship, and manufacturing techniques to insure that the functional capabilities of the jack will meet the design objective upon which the drawing is based.

4.2 Acceptance test. The acceptance test shall consist of:

a. Individual tests

b. Sampling test

4.2.1 Individual tests. The jack shall be subjected to the following tests as described under 4.3:

a. Examination of product

b. Leakage test

c. By-pass valve test

* 4.2.2 Sampling tests. One jack shall be selected from each 25 or fraction thereof on the contract or order and subjected to the static proof test described in 4.3.4. Items selected shall have successfully passed all individual tests.

4.3 TEST METHODS

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4.3.1 Examination of product. The jack shall be inspected to determine compliance with the requirements specified herein with respect to workmanship, marking, and compliance with the drawing.

4.3.2 Leakage test. The cylinder and ram assembly of the assembled jack shall be tested by applying a minimum hydraulic pressure of 7,500 psi. This test shall be performed with an appropriate load on the ram assembly to preclude damage to ram stops. Any leaking of hydraulic fluid shall be cause for rejection.

4.3.3 By-pass valve test. The pump shall be tested by sealing the pump outlet and operating the pump until the by-pass valve opens. The by-pass valve shall be adjusted to the valve specified on the drawing.

4.3.4 Static proof test. The jack, after the by-pass valve has been adjusted, shall be tested by raising a 30-ton load vertically from the lowest position to the fully extended position. At the fully extended position, under a 20-ton load, a horizontal load of 3 tons shall be applied to the top of the extended screw extension. Failure of any part of the jack shall be cause for rejection.

4.3.5 Low temperature test. The jack assembly shall be placed in a test chamber and cooled to -54° C for a period of 72 hours. At the end of this period and at -54° C, the jack ram shall be raised to its maximum extension and pumped until the by-pass relief valve operates. The lowering valve shall be opened and the ram shall retract with not more than 50 pounds applied to the ram. For this test the jack shall be serviced with hydraulic fluid comforming to MIL-H-81019. The number of full strokes necessary to fully extend the ram shall be recorded at both $+21^{\circ}$ C and -54° C. The strokes required to fully extend the ram at -54° C shall not be more than double the strokes required at $+21^{\circ}$ C.

4.3.6 Life test. The life test shall consist of 200 cycles of raising a 20-ton load vertically and 50 cycles of raising a 20-ton load vertically with a horizontal load of 3 tons applied to the top of the extended screw extension. One cycle shall consist of raising the ram from collapsed height to the fully extended position and return to the collapsed position. The horizontal load shall be graduated from 0 in the collapsed position to 3 tons in the fully extended position. Failure of any part of the jack during this test shall be cause for rejection.

* 4.4 FIRST ARTICLE TESTING

* 4.4.1 First Article test sample. The first article test sample shall consist of one jack representative of the production jack.

* 4.4.2 Test report. After completion of the first article tests, a test report shall be prepared in accordance with MIL-STD-831 and three complete copies furnished the procuring activity.

* 4.4.3 First Article tests. First article tests shall consist of all tests described under 4.3.

* 5. PACKAGING

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

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* 5.1.1 Level A. The jack shall be preserved in accordance with MIL-P-116, Method I.

* 5.1.2 Level C. The jack shall be preserved and packaged in accordance with the Manufacturer's commercial practice.

* 5.2 Packing. Packing shall be Level A, B or C as specified (see 6.2)

* 5.2.1 Level A. The jack preserved and packaged to meet 5.1.1 shall be packed in a container conforming to PPP-B-601, overseas type. The closure of shipping container shall be in accordance with the Appendix of the shipping container specification.

* 5.2.2 Level B. Same as Level A except the container shall be domestic type.

* 5.2.3 Level C. Each jack preserved and packaged as specified in 5.1.1 shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with the Uniform Freight Classification Rules or regulations of other carrier as applicable to the mode of transportation.

* 5.3 Physical protection. The jack shall be cushioned, anchored, blocked and braced in accordance with MIL-STD-1186. The Freefall Drop Test, Edgewise Drop Test and Cornerwise Test shall be in accordance with Appendix A of MIL-STD-1186. The Tests are not required when Level C packing is specified.

* 5.4 Marking. In addition to any special marking required by the contract or order (see 6.3) interior packages and exterior containers shall be marked in accordance with MIL-STD-129.

5.4.1 Reinspection date marking. Reinspection marking shall be in accordance with Bulletin 405.

6. Notes

6.1 Intended use. The type F-1 jack covered by this specification is intended for use as an aircraft axle jack.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

* a. Title, number, and date of this specification.

* b. Selection of applicable levels of preservation, packaging, and packing (see section 5)

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* 6.2.2 First Article tests. One jack will be required as a first article sample to be subjected to the preproduction first article tests to determine compliance with the requirements of this specification. The point of inspection for these tests will be specified.

6.2.3 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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